

Based on the project details provided for the **LED Display** task, here is a guide to help you build your programmable LED matrix system using an Arduino.

Project Overview

- **Objective:** To create a programmable display capable of showing messages via an LED matrix.
- **Complexity:** Easy.
- **Estimated Duration:** 5 hours.
- **Real-World Application:** Digital signage and electronic notice boards.

Components Needed

- **Arduino Board:** (Uno or Nano are ideal for this "Easy" complexity level).
- **LED Matrix:** Commonly a **MAX7219** 8x8 matrix module, which simplifies wiring.
- **Power Supply:** Standard USB power or a 5V adapter.
- **Jumper Wires:** For connecting the matrix to the Arduino.

Wiring Guide (Using MAX7219)

The MAX7219 module uses the SPI communication protocol to interface with your Arduino.

LED Matrix Pin	Arduino Pin (Typical)
VCC	5V
GND	GND
DIN (Data In)	Pin 11 (MOSI)
CS (Load)	Pin 10 (SS)
CLK (Clock)	Pin 13 (SCK)

Basic C++ Code Structure

You can use the **LedControl** or **MD_Parola** libraries to handle the data display and matrix programming.

```
#include <LedControl.h>
```

```
// Pin setup: DIN, CLK, CS, Number of Devices
LedControl lc = LedControl(11, 13, 10, 1);
```

```
void setup() {
    lc.shutdown(0, false);           // Wake up the MAX7219
    lc.setIntensity(0, 8);           // Set brightness (0-15)
    lc.clearDisplay(0);              // Clear the matrix
}
```

```
void loop() {
    // Example: Display a simple character or scroll text
    // The logic here focuses on LED matrix programming and data display
}
```

Learning Outcomes

By completing this task, you will gain hands-on experience in:

- **LED Matrix Programming:** Learning how to address individual pixels or rows/columns to form characters.
- **Data Display:** Understanding how to translate string data into visual patterns on hardware.